

IMO Training 2019-20 (Phase III)
Number Theory Problem Set

1. Find all $m, n \in \mathbb{Z}$ s.t. $7^m = 5^n + 24$.
2. Find all $N \in \mathbb{N}$ for which N has at least 4 divisors, and the sum of squares of the 4 smallest positive divisors of N equals N .
3. Find all $a, b, c \in \mathbb{N}$ s.t. $a^2 + b + 3 = (b^2 - c^2)^2$.
4. Find all $m, n \in \mathbb{Z}$ s.t. $\frac{n^2 + 1}{2m}$ and $\sqrt{2^{n-1} + m + 4}$ are both integers.
5. Find all integers $n > 1$ and $a_1, a_2, \dots, a_n \in \mathbb{Z}$ s.t.
 - (i) $2 < a_1 \leq a_2 \leq \dots \leq a_n$;
 - (ii) $a_1, a_2, \dots, a_n$ are divisors of $15^{25} + 1$;
 - (iii) $2 - \frac{2}{15^{25} + 1} = \left(1 - \frac{2}{a_1}\right) + \left(1 - \frac{2}{a_2}\right) + \dots + \left(1 - \frac{2}{a_n}\right)$
6. Given integer $n \geq 2$, show that $\exists a, b \in \mathbb{Z}$ s.t. $\forall m \in \mathbb{Z}$, n does not divide $m^3 + am + b$.